Some Fourier transform theory

Integral Master

May 2026

1 Abstract

In this following document, we present an expository paper on the Fourier transform, some of its properties, its inversion, its relation with a variety of functional spaces, and end with presenting the uses of the theorems and properties in physics.

2 Definitions of the Fourier transform:

Definition 2.1. *Let $f : \mathbb{R}^n \to \mathbb{C}$, $f \in L^1(\mathbb{R}^n)$, and λ denote the Lebesgue measure on $\mathbb{R}^n$ then we have the Fourier transform of f:*

$$\mathcal{F}\{f(x)\}(\xi) = \hat{f}(\xi) = \int_{\mathbb{R}^n} f(x)e^{-2\pi i\xi\cdot x}d\lambda(x).$$

It is evident why the Fourier transform is well defined as a consequence of $f \in L^1(\mathbb{R}^n)$:

$$\left|\int_{\mathbb{R}^n} f(x)e^{-2\pi i\xi\cdot x}dx\right| \leq \int_{\mathbb{R}^n} |f(x)|\,dx, (|e^{-2\pi i\xi\cdot x}| = 1).$$

Of course, this is not the only convention for the Fourier transform:

Definition 2.2.

$$\hat{f}(\omega) = \int_{\mathbb{R}^n} f(x)e^{-i\omega\cdot x}d\lambda(x),$$

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}^n}\int_{\mathbb{R}^n} f(x)e^{-i\omega\cdot x}d\lambda(x).$$

$$\hat{\psi}(p,t) = \frac{1}{\sqrt{2\pi\hbar}^n}\int_{\mathbb{R}^n} \psi(x,t)e^{-\frac{ip\cdot x}{\hbar}}d\lambda(x), (p = \hbar k)$$

The last one is actual the momentum-space Fourier transform, which is imperative in Quantum theory (though it was most probably made obvious by the presence of the Planck constant.)

3 Certain properties of the Fourier transform:

Theorem 3.1 (The scaling property of the Fourier transform)**.** *For a function $f \in L^1(\mathbb{R}^n)$ and $a \in \mathbb{R} \setminus \{0\}$ we have:*

$$\mathcal{F}\{f(ax)\}(\xi) = \frac{1}{a^n}\hat{f}\left(\frac{\xi}{a}\right)$$

Proof. Let $f \in L^1(\mathbb{R}^n)$ and $a \in \mathbb{R} \setminus \{0\}$, we must first prove that the function $f(ax) \in L^1(\mathbb{R}^n)$:

$$\|f(ax)\|_1 = \int_{\mathbb{R}^n} |f(ax)| d\lambda(x) = \int_{\mathbb{R}^n} |f(x)| \cdot \frac{1}{a^n} d\lambda(x) = \frac{1}{a^n}\|f\|_1 < \infty \implies f(ax) \in L^1(\mathbb{R}^n).$$

$$\mathcal{F}\{f(ax)\}(\xi) = \int_{\mathbb{R}^\ltimes} f(ax)e^{-2\pi i\xi\cdot x} d\lambda(x) = \int_{\mathbb{R}^n} f(x)e^{-2\pi i\xi\cdot\left(\frac{x}{a}\right)} \cdot \frac{1}{a^n} d\lambda(x)$$

$$= \frac{1}{a^n}\int_{\mathbb{R}^n} f(x)e^{-2\pi i\left(\frac{\xi}{a}\right)\cdot x} d\lambda(x) = \frac{1}{a^n}\hat{f}\left(\frac{\xi}{a}\right)$$

□

Theorem 3.2. *A subset of the set of eigenfunctions of the Fourier transform over $\mathbb{R}^n$ is the set:*

$$\{e^{-a\|x\|^2} \big| \Re(a) > 0\}$$

.

Proof. Let us first compute the Fourier transform of $e^{-\pi\|x\|^2}$ (assuming $x = (x_1, \ldots, x_n)$ and $\xi = (\xi_1, \ldots, \xi_n)$):

$$\mathcal{F}\{e^{-\pi\|x\|^2}\}(\xi) = \int_{\mathbb{R}^n} e^{-\pi\|x\|^2} e^{-2\pi i\xi\cdot x} d\lambda(x) = \int_{\mathbb{R}^n} e^{-\pi\|x\|^2 - 2\pi i\xi\cdot x} d\lambda(x)$$

$$= \int_{\mathbb{R}^n} e^{-\pi\sum_{k=1}^n x_k^2 - 2\pi i\sum_{k=1}^n \xi_k x_k} d\lambda(x_1, \ldots, x_n) = \int_{\mathbb{R}^n} e^{-\pi\cdot\sum_{k=1}^n \left(x_k^2 - 2i\xi_k x_k\right)} d\lambda(x_1, \ldots, x_n)$$

$$= \int_{\mathbb{R}^n} \prod_{k=1}^n e^{-\pi\left(x_k^2 - 2i\xi_k x_k\right)} d\lambda(x_1, \ldots, x_n) = \prod_{k=1}^n \int_{-\infty}^{\infty} e^{-\pi\left(x_k^2 - 2i\xi_k x_k\right)} d\lambda(x_k)$$

$$= \prod_{k=1}^n \int_{-\infty}^{\infty} e^{-\pi(x_k^2 - 2i\xi_k x_k + (i\xi_k)^2 - (i\xi_k)^2)} d\lambda(x_k) = \prod_{k=1}^n \int_{-\infty}^{\infty} e^{-\pi(x_k - i\xi_k)^2} e^{-\pi(-i\xi_k)^2} d\lambda(x_k)$$

$$= \prod_{k=1}^n e^{-\pi\xi_k^2} \int_{-\infty}^{\infty} e^{-\pi(x_k - i\xi_k)^2} d\lambda(x_k) = e^{-\pi\sum_{k=1}^n \xi_k^2} \int_{-\infty}^{\infty} e^{-\pi(x_k - i\xi_k)^2} d\lambda(x_k)$$

$$= e^{-\pi\|\xi\|^2} \prod_{k=1}^n \int_{-\infty+i\xi_k}^{\infty+ik} e^{-\pi x_k^2} d\lambda(x_k).$$

We have by Cauchy's integral theorem that $\int_{-\infty+i\xi_k}^{\infty+i\xi_k} e^{-\pi x_k^2} d\lambda(x_k) = \int_{-\infty}^{\infty} e^{-\pi x_k^2} d\lambda(x_k) = 1$

$$\implies \mathcal{F}\{e^{-\pi\|x\|^2}\}(\xi) = e^{-\pi\|\xi_k\|^2}$$

And by the scaling property:

$$\mathcal{F}\{e^{-\pi a\|x\|^2}\}(\xi) = \frac{1}{\sqrt{a}^n} e^{-\pi \frac{\|\xi\|^2}{a}}.$$

□

Lemma 3.3 (The Fourier transform, differentiation, and multiplication)**.** *Let $f \in L^1(\mathbb{R}^n)$, let $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$ and $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{N}^n$.*

We write $x^\alpha = x_1^{\alpha_1} \cdot x_2^{\alpha_2} \cdot \cdots \cdot x_n^{\alpha_n}$ and $\partial_x^\alpha = \frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \ldots \partial_{x_n}^{\alpha_n}}$.

We may conclude the following results:

$$\mathcal{F}\{x^\alpha f(x)\}(\xi) = \left(-\frac{1}{2\pi i}\right)^{|\alpha|} \partial_\xi^\alpha \hat{f}(\xi)$$

if $x^\alpha f(x) \in L^1(\mathbb{R}^n)$, and:

$$\mathcal{F}\{\partial_x^\alpha f(x)\}(\xi) = (2\pi i)^{|\alpha|} \xi^\alpha \hat{f}(\xi)$$

if $\partial_x^\alpha f(x) \in L^1(\mathbb{R}^n)$

Proof. Suppose $f \in L^1(\mathbb{R}^n)$ with $x^\alpha f(x) \in L^1(\mathbb{R}^n)$

$$\mathcal{F}\{x^\alpha f(x)\}(\xi) = \int_{\mathbb{R}^n} x^\alpha f(x) e^{-2\pi i \xi \cdot x} d\lambda(x)$$

$$= \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} x_1^{\alpha_1} \ldots x_n^{\alpha_n} f(x_1, \ldots, x_n) e^{-2\pi i \xi_1 x_1} \ldots e^{-2\pi i \xi_n x_n} d\lambda(x)$$

$$= \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x_1, \ldots, x_n) \left(-\frac{1}{2\pi i} \frac{\partial^{\alpha_1}}{\partial \xi_1^{\alpha_1}} e^{-2\pi i \xi_1 x_1}\right) \ldots \left(-\frac{1}{2\pi i} \frac{\partial^{\alpha_n}}{\partial \xi_n^{\alpha_n}} e^{-2\pi i \xi_n x_n}\right) d\lambda(x)$$

$$= \left(-\frac{1}{2\pi i}\right)^{|\alpha|} \int_{\mathbb{R}^n} f(x) \partial_\xi^\alpha (e^{-2\pi i \xi \cdot x}) d\lambda(x) = \left(-\frac{1}{2\pi i}\right)^{|\alpha|} \int_{\mathbb{R}^n} \partial_\xi^\alpha (f(x) e^{-2\pi i \xi \cdot x}) d\lambda(x)$$

$$= \left(-\frac{1}{2\pi i}\right)^{|\alpha|} \partial_\xi^\alpha \int_{\mathbb{R}^n} f(x) e^{-2\pi i \xi \cdot x} d\lambda(x) = \left(-\frac{1}{2\pi i}\right)^{|\alpha|} \partial_\xi^\alpha \hat{f}(\xi)$$

Suppose $f \in L^1(\mathbb{R}^n)$ with $\partial_x^\alpha f(x) \in L^1(\mathbb{R}^n)$

$$\mathcal{F}\{\partial_x^\alpha f(x)\}(\xi) = \int_{\mathbb{R}^n} \partial_x^\alpha f(x) e^{-2\pi i \xi \cdot x} d\lambda(x)$$

$$= \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \frac{\partial^{\alpha_1}}{\partial x_1^{\alpha_1}} \cdots \frac{\partial^{\alpha_n}}{\partial x_n^{\alpha_n}} f(x_1, \ldots, x_n) e^{-2\pi i \xi_1 x_1} \ldots e^{-2\pi i \xi_n x_n} d\lambda(x_1, \ldots, x_n)$$

We now proceed to use integration by parts in each coordinate, and assume $\frac{\partial^j f}{\partial x_j^n} \to 0$ as $|x_k| \to \infty, \forall k, \forall j \in \{1, \ldots, \alpha_k\}$.

$$\int_{-\infty}^{\infty} \frac{\partial^{\alpha_k} f}{\partial x_k^{\alpha_k}} e^{-2\pi i \xi_k x_k} d\lambda(x_k) = \left[\frac{\partial^{\alpha_k - 1} f}{\partial x_k^{\alpha_k - 1}} e^{-2\pi i \xi_k x_k}\right]_{-\infty}^{\infty} - (-2\pi i \xi_k) \int_{-\infty}^{\infty} \frac{\partial^{\alpha_k - 1} f}{\partial x_k^{\alpha_k - 1}} e^{-2\pi i \xi_k x_k} d\lambda(x_k)$$

$$= 2\pi i\xi_k \int_{-\infty}^{\infty} \frac{\partial^{\alpha_k - 1} f}{\partial x_k^{\alpha_k - 1}} e^{-2\pi i \xi_k x_k} d\lambda(x_k)$$

$$= 2\pi i\xi_k \left(2\pi i\xi_k \int_{-\infty}^{\infty} \frac{\partial^{\alpha_k - 2} f}{\partial x_k^{\alpha_k - 2}} e^{-2\pi i \xi_k x_k} d\lambda(x_k) \right) = (2\pi i\xi_k)^2 \int_{-\infty}^{\infty} \frac{\partial^{\alpha_k - 2} f}{\partial x_k^{\alpha_k - 2}} e^{-2\pi i \xi_k x_k} d\lambda(x_k).$$

Applying this recursively:

$$\int_{-\infty}^{\infty} \frac{\partial^{\alpha_k} f}{\partial x_k^{\alpha_k}} e^{-2\pi i \xi_k x_k} d\lambda(x_k) = (2\pi i\xi_k)^{\alpha_k} \int_{-\infty}^{\infty} f(x_1, \ldots, x_k, \ldots, x_n) e^{-2\pi i \xi_k x_k} d\lambda(x_k).$$

In sum, we obtain the following result:

$$\mathcal{F}\{\partial_x^\alpha f(x)\}(\xi) = \left(\prod_{k=1}^{n} (2\pi i\xi_k)^{\alpha_k} \right) \hat{f}(\xi) = (2\pi i)^{\sum_{k=1}^n \alpha_k} \left(\prod_{k=1}^{n} \xi_k^{\alpha_k} \right) \hat{f}(\xi)$$

$$= (2\pi i)^{|\alpha|} \xi^\alpha \hat{f}(\xi).$$

□

4 The Fourier inversion theorem:

Naturally, we have to discuss inversion of the Fourier transform, allowing us to obtain a function from a Fourier transform, an imperative step in solving many differential equations with the Fourier transform, which leads us to the next topic:

Theorem 4.1 (The Fourier inversion theorem)**.** *Given $f \in L^1(\mathbb{R}^n)$ with $\hat{f} \in L^1(\mathbb{R}^n)$, and assuming the first convention, we have:*

$$\int_{\mathbb{R}^n} \hat{f}(\xi) e^{2\pi i \xi \cdot x} dx = f(x).$$

Proof. Let $f \in L^1(\mathbb{R}^n)$ with $\hat{f} \in L^1(\mathbb{R}^n)$.

$$\hat{f}(\xi) = \int_{\mathbb{R}^n} f(x) e^{-2\pi i \xi \cdot x} d\lambda(x)$$

$$\implies \hat{f}(\xi) e^{-2\pi i \xi \cdot y} = \int_{\mathbb{R}^n} f(x) e^{-2\pi i \xi \cdot (y-x)} d\lambda(x).$$

We use Gaussian regularization using the function $e^{-\pi\epsilon^2|\xi|^2}$:

$$\hat{f}(\xi) e^{-2\pi i \xi \cdot y} e^{-\pi\epsilon^2|\xi|^2} = \int_{\mathbb{R}^n} f(x) e^{-2\pi i \xi \cdot (y-x)} e^{-\pi\epsilon^2\|\xi\|^2} d\lambda(x)$$

$$\implies \int_{\mathbb{R}^\kappa} \hat{f}(\xi) e^{-2\pi i \xi \cdot y} e^{-\pi\epsilon^2\|\xi\|} d\lambda(\xi) = \int_{\mathbb{R}^n} \int_{\mathbb{R}^\kappa} f(x) e^{-2\pi i \xi \cdot (y-x)} e^{-\pi\epsilon^2\|\xi\|^2} d\lambda(x) d\lambda(\xi)$$

by Fubini's theorem:

$$\int_{\mathbb{R}^n} \int_{\mathbb{R}^\kappa} f(x) e^{-2\pi i \xi \cdot (y-x)} e^{-\pi\epsilon^2\|\xi\|^2} d\lambda(x) d\lambda(\xi) = \int_{\mathbb{R}^n} f(x) \left(\int_{\mathbb{R}^n} e^{-\pi\epsilon^2\|\xi\|^2} e^{-2\pi i \xi \cdot (y-x)} d\lambda(\xi) \right) d\lambda(x)$$

$$= \int_{\mathbb{R}^n} f(x) \frac{1}{\epsilon^n} e^{-\pi \frac{\|y-x\|^2}{\epsilon^2}} d\lambda(x)$$

Using the substitution:

$$x = \Phi(u) = y - \epsilon u \implies \det(J_\Phi) = \epsilon^n$$

$$\implies \int_{\mathbb{R}^n} f(x) \frac{1}{\epsilon^n} e^{-\pi \frac{\|y-x\|^2}{\epsilon^2}} d\lambda(x) = \int_{\mathbb{R}^n} f(y-\epsilon u) \frac{1}{\epsilon^n} e^{-\pi \|u\|^2} \epsilon^n d\lambda(u) = \int_{\mathbb{R}^n} f(y-\epsilon u) e^{-\pi \|u\|^2} d\lambda(u).$$

$$|f(y-\epsilon u)| \leq \|f\|_\infty = \sup_{x \in \mathbb{R}^n} |f(x)| \implies \left| f(y-\epsilon u) e^{-\pi \|u\|^2} \right| \leq \|f\|_\infty e^{-\pi \|u\|} \in L^1(\mathbb{R}^n)$$

and by the dominated convergence theorem:

$$\lim_{\epsilon \to 0^+} \int_{\mathbb{R}^n} f(y-\epsilon u) e^{-\pi \|u\|^2} d\lambda(u) = \int_{\mathbb{R}^n} \lim_{\epsilon \to 0^+} f(y-\epsilon u) e^{-\pi \|u\|^2} d\lambda(u) = f(y) \int_{\mathbb{R}^n} e^{-\pi \|u\|^2} d\lambda(u) = f(y)$$

Since $\hat{f} \in L^1(\mathbb{R}^n)$, we have:

$$\left| \hat{f}(\xi) e^{-2\pi i \xi \cdot y} e^{-\pi \epsilon^2 \|\xi\|} \right| \leq \left| \hat{f}(\xi) \right| \in L^1(\mathbb{R}^n).$$

By the dominated convergence theorem:

$$\lim_{\epsilon \to 0^+} \int_{\mathbb{R}^n} \hat{f}(\xi) e^{-\pi \epsilon^2 \|\xi\|^2} e^{-2\pi i \xi \cdot y} d\lambda(\xi) = \int_{\mathbb{R}^n} \lim_{\epsilon \to 0^+} \hat{f}(\xi) e^{-\pi \epsilon^2 \|\xi\|^2} e^{-2\pi i \xi \cdot x} d\lambda(\xi)$$

$$= \int_{\mathbb{R}^n} \hat{f}(\xi) e^{-2\pi i \xi \cdot y} d\lambda(y)$$

$$\therefore f(y) = \int_{\mathbb{R}^n} \hat{f}(\xi) e^{-2\pi i \xi \cdot y} d\lambda(y).$$

Thus, the proof is concluded. □

5 The Fourier transform and the L^2 inner product, Plancherel's theorem.

Definition 5.1 (The L^2 inner product)**.** *Let $(X, \mathcal{A}, \mu)$ be a measure space. The Mathematics convention for the L^2 inner product is defined as follows:*

$$\forall f, g \in L^2(X, \mu) : \langle f, g \rangle_{L^2} = \int_X f(x) \overline{g(x)} d\mu(x)$$

and:

$$\forall f \in L^2(X, \mu) : \|f\|_2^2 = \langle f, f \rangle$$

The Physics convention for the L^2 inner product is defined as follows:

$$\forall \psi, \varphi \in L^2(X, \mu) : \langle \psi | \phi \rangle = \int_X \overline{\psi(x)} \phi(x) d\mu(x)$$

and:

$$\forall \psi \in L^2(X, \mu) : \|\psi\|_2^2 = \langle \psi | \psi \rangle.$$

We will proceed with the Mathematics convention, the Physics convention does not change much.

Theorem 5.1 (Self-adjoint-ness of the Fourier transform)**.** *The Fourier transform operator $\mathcal{F}$ is self-adjoint in the L^2 inner product, that is:*

$$\forall f, g \in L^1(\mathbb{R}^n) \bigcap L^2(\mathbb{R}^n), \hat{f} \in L^1(\mathbb{R}^n) \wedge \hat{g} \in L^1(\mathbb{R}^n) : \langle \hat{f}, g \rangle_{L^2} = \langle f, \hat{g} \rangle_{L^2}.$$

Proof. Let $f, g \in L^1(\mathbb{R}^n) \bigcap L^2(\mathbb{R}^n)$ with $\hat{f} \in L^1(\mathbb{R}^n) \wedge \hat{g} \in L^1(\mathbb{R}^n)$ (to allow using The Fourier inversion theorem.)

$$\langle \hat{f}, g \rangle_{L^2} = \int_{\mathbb{R}^n} \hat{f}(\xi) \overline{g(\xi)} d\lambda(\xi) = \int_{\mathbb{R}^\ltimes} \hat{f}(\xi) \overline{\left(\int_{\mathbb{R}^n} \hat{g}(x) e^{-2\pi i \xi \cdot x} d\lambda(x) \right)} d\lambda(\xi)$$

$$= \int_{\mathbb{R}^n} \hat{f}(\xi) \left(\int_{\mathbb{R}^n} \overline{\hat{g}(x)} e^{2\pi i \xi \cdot x} d\lambda(x) \right) d\lambda(\xi) = \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \hat{f}(\xi) \overline{\hat{g}(x)} e^{2\pi i \xi \cdot x} d\lambda(x) d\lambda(\xi).$$

By Fubini's theorem:

$$\int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \hat{f}(\xi) \overline{\hat{g}(x)} e^{2\pi i \xi \cdot x} d\lambda(x) d\lambda(\xi) = \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \hat{f}(\xi) \overline{\hat{g}(x)} e^{2\pi i \xi \cdot x} d\lambda(\xi) d\lambda(x)$$

$$= \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} \hat{f}(\xi) e^{2\pi i \xi \cdot x} d\lambda(\xi) \right) \overline{\hat{g}(x)} d\lambda(x) = \int_{\mathbb{R}^n} f(x) \overline{\hat{g}(x)} d\lambda(x) = \langle f, \hat{g} \rangle_{L^2}.$$

Thus, the proof is concluded. □

Theorem 5.2.

$$\forall f, g \in L^1(\mathbb{R}^n) \bigcap L^2(\mathbb{R}^n), \hat{f}, \hat{g} \in L^1(\mathbb{R}^n) : \langle f, g \rangle_{L^2} = \langle \hat{f}, \hat{g} \rangle_{L^2}.$$

Proof. Let $f, g \in L^1(\mathbb{R}^n) \bigcap L^2(\mathbb{R}^n)$ with $\hat{f}, \hat{g} \in L^1(\mathbb{R}^n)$

$$\langle f, g \rangle_{L^2} = \int_{\mathbb{R}^n} f(x) \overline{g(x)} d\lambda(x)$$

$$= \int_{\mathbb{R}^n} f(x) \overline{\left(\int_{\mathbb{R}^n} \hat{g}(\xi) e^{2\pi i \xi \cdot x} d\lambda(\xi) \right)} d\lambda(x)$$

$$= \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} f(x) \overline{\hat{g}(\xi)} e^{-2\pi i \xi \cdot x} d\lambda(x)$$

$$= \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} f(x) e^{-2\pi i \xi \cdot x} d\lambda(x) \right) \overline{\hat{g}(\xi)} d\lambda(\xi) = \int_{\mathbb{R}^n} \hat{f}(\xi) \overline{\hat{g}(\xi)} d\lambda(\xi)$$

$$= \langle \hat{f}, \hat{g} \rangle_{L^2}.$$

Thus, the proof is concluded. □

Theorem 5.3 (Plancherel's theorem/Rayleigh's theorem/Rayleigh's energy theorem.)**.** *For any function $f \in L^1(\mathbb{R}^n) \bigcap L^2(\mathbb{R}^n)$ with $\hat{f} \in L^1(\mathbb{R}^n)$*

Proof. Let $f \in L^1(\mathbb{R}^n) \bigcap L^2(\mathbb{R}^n)$ with $\hat{f} \in L^1(\mathbb{R}^n)$.

By Theorem 4.2:

$$\|\hat{f}\|_2^2 = \langle \hat{f}, \hat{f} \rangle_{L^2} = \langle f, f \rangle_{L^2} = \|f\|_2^2$$
$$\implies \|f\|_2^2 = \|\hat{f}\|_2^2$$

Thus, the proof is concluded. □

Fun fact 6. *Theorem 4.2 is at times called "Rayleigh's theorem" or "Rayleigh's energy theorem" due to its original formulation by Lord John William Strutt, 3rd Baron Rayleigh (1889), a British physicist, who gave the physical interpretation of the earlier identity as follows:*

The total energy of a signal in the time domain

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

is the same as the total energy of its Fourier transform in the frequency domain

$$E = \int_{-\infty}^{\infty} |\hat{x}(\xi)|^2 d\xi.$$

Corollary 6.0.1. *The Fourier transform is an isometry on L^2, for if we take f and g as in Theorem 4.2, we have:*

$$\|f - g\|_2 = \|\mathcal{F}\{f - g\}\|_2 = \|\mathcal{F}f - \mathcal{F}g\|_2$$

7 The Schwarz space:

Definition 7.1 (The Schwarz space)**.** *Let $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$ and $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{N}^n$, we write $x^\alpha = x_1^{\alpha_1} \cdot x_2^{\alpha_2} \cdot \cdots \cdot x_n^{\alpha_n}$ and $\partial_x^\alpha = \frac{\partial^{|\alpha|}}{\partial {x_1}^{\alpha_1} \ldots \partial {x_n}^{\alpha_n}}$.*

$$\mathcal{S}(\mathbb{R}^n) = \left\{ f \in \mathcal{C}^\infty(\mathbb{R}^n) \middle| \forall \alpha, \beta \in \mathbb{N}^n \|x^\alpha \partial_x^\beta f\|_\infty = \sup_{x \in \mathbb{R}^n} |x^\alpha \partial_x^\beta f| < \infty \right\}.$$

Theorem 7.1 (The Fourier transform on the Schwarz space)**.** *The Fourier transform operator $\mathcal{F}$ is an automorphism on $\mathcal{S}(\mathbb{R}^n)$, that is:*

$$\mathcal{F} : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n), \mathcal{F}^{-1} : \mathcal{S}(\mathbb{R}^\bowtie) \to \mathcal{S}(\mathbb{R}^n)$$

where $\mathcal{F}^{-1}\{\hat{f}(\xi)\}(x) = \int_{\mathbb{R}^n} \hat{f}(\xi) e^{2\pi i \xi \cdot x} d\lambda(\xi)$. And since $\mathcal{S}(\mathbb{R}^n) \subset L^p(\mathbb{R}^n), \forall p \in [1, \infty]$, $\mathcal{F}$ and $\mathcal{F}^{-1}$ are always defined and Plancherel's theorem always holds.

Proof. It suffices to prove that the Fourier transform operator $\mathcal{F}$ maps from $\mathcal{S}(\mathbb{R}^n)$ to itself. By the earlier theorems:

$$\mathcal{F}\{\partial_x^\beta f(x)\}(\xi) = (2\pi i)^{|\beta|} \xi^\beta \hat{f}(\xi)$$

and

$$\mathcal{F}\{x^\alpha f(x)\}(\xi) = \left(-\frac{1}{2\pi i} \right)$$

□